

Generalization-Aware Safety Monitoring for Surgical Segmentation Failures under Acquisition Degradation

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Abstract. Surgical segmentation networks can fail silently under acquisition degradation: the mask is wrong, yet the network remains confident. We present TCSR-Monitor, a generalization-aware safety-monitoring framework that maps 22 observable confidence, shape, temporal, and image-quality features from a frozen segmenter to a failure-risk score. The contribution is not only a detector, but a validity protocol for asking whether learned alarms remain credible under distribution shift. On EndoVis 2017 instrument segmentation, TCSR-Monitor attains AUROC 0.814 on unseen corruption types under leave-one-corruption-out (LOCO) evaluation, winning 26 of 30 held-out conditions against entropy. A circularity-control experiment shows that the monitor predicts segmentation failure rather than merely detecting corruption (within-corrupted AUROC 0.946, [0.943, 0.949]). A learned confidence-only control is strong, but the full feature set improves LOCO AUROC/AUPRC at $\tau = 0.75$ from 0.912/0.888 to 0.920/0.903. Finally, Mondrian group-conditional conformal calibration restores near-target per-severity miss-rate at $\alpha = 0.10$ (severity-1 miss 50.5% $\rightarrow$ 9.3%), while making the increased frame-level alarm burden explicit. Zero-shot transfer to SAM2-Tiny reaches AUROC 0.930 [0.912, 0.945], supporting feature portability but also showing that entropy remains strong when failures are already expressed as uncertainty.

Keywords: Surgical segmentation · Failure monitoring · Conformal prediction

1 Introduction

Deep segmentation networks are increasingly used to parse endoscopic video into instrument and anatomy masks [1,2], but deployed surgical video contains smoke, defocus, illumination change, compression, and motion blur that are under-represented in curated benchmarks. These acquisition shifts can produce *silent failures*: masks with low intersection-over-union (IoU) despite high output confidence. Such failures are safety-critical because standard confidence monitors—max-softmax [3], predictive entropy [5], and calibration-based scores [6]—measure uncertainty. If a wrong mask is also confident, uncertainty is low by construction.

TCSR-Monitor addresses this gap by learning observable failure patterns rather than directly reading uncertainty. It wraps a frozen segmenter, extracts confidence, morphology, temporal-consistency, and image-quality features, and predicts whether the segmenter has failed without accessing ground truth at deployment. A learned monitor, however, is useful only if it survives three validity tests: it must generalize beyond seen degradations, predict failure rather than corruption presence, and support calibrated alarm thresholds across degradation severity. This paper is organized around those tests.

Contributions.

1. a lightweight failure monitor for frozen surgical segmenters using observable confidence, shape, temporal, and image-quality features;
2. a protocol-qualified generalization hierarchy showing AUROC 0.814 on unseen corruption types under LOCO evaluation;
3. a circularity control showing that the monitor predicts failure rather than corruption presence (within-corrupted AUROC 0.946);
4. Mondrian conformal calibration that reduces severity-1 miss-rate from 50.5% to 9.3% at $\alpha = 0.10$, while exposing the false-alarm cost;
5. supporting controls, reported in the supplement, that bound the claim against confidence-only, learner, temporal-order, sequence-held-out, and SAM2 transfer alternatives.

The intended use is not to replace segmentation, but to add a fast, external watchdog for cases in which the segmenter remains confident while its mask quality collapses. This distinction also determines the evaluation: a monitor that performs well only after seeing the same corruption operators during training is less useful than one whose failure score transfers across unseen degradation types.

2 Related Work

Instrument segmentation has advanced through U-Net variants [7], transformers [8], and foundation models such as SAM/SAM2 [9,10]. Robustness benchmarks expose sensitivity to common or surgical corruptions [4,11,13], but they primarily evaluate the segmenter itself. Our focus is deployment-time monitoring after a mask has already been produced.

Confidence scores, entropy, calibration, test-time ensembles, and feature-space OOD scores are common post-hoc reliability signals [3,5,6,14,15]. In medical segmentation, however, uncertainty can be calibrated in aggregate while still failing on individual cases [18], and networks may remain overconfident on wrong masks [19]. Learned failure prediction therefore provides a complementary route: prior work estimates model confidence or segment quality from auxiliary signals [17,16,20,21]. We differ by centering surgical acquisition degradation, held-out corruption protocols, and a circularity control that separates failure prediction from corruption detection. Our alarm layer relates to selective prediction [24,25] and conformal risk control [22,23,26]; group-conditional Mondrian calibration [27] is especially natural when degradation severity defines safety-relevant groups.

Frame I_t	→	Frozen segmenter	→	Observable features	→	Risk score	→	Alarm
raw image		$p_t, \hat{y}_t$		confidence, shape, temporal, quality		$g_\phi(x_t)$		$g_\phi(x_t) > \lambda_g$

Fig. 1. Overview of TCSR-Monitor. The monitor observes the raw frame and the segmenter’s output, but does not modify the segmenter or use ground truth at deployment. Calibration converts risk scores into global or group-specific alarm thresholds.

3 Method

TCSR-Monitor wraps a frozen segmenter f_θ without changing its weights. For each frame I_t , the segmenter produces a probability map p_t and mask $\hat{y}_t$. A frame is labeled a failure during training/evaluation when

$$\text{fail}(I_t) = \mathbf{1}[\text{IoU}(\hat{y}_t, y_t^*) < \tau], \quad \tau \in \{0.5, 0.75\}. \quad (1)$$

The monitor extracts 22 scalar features from p_t , $\hat{y}_t$, I_t , and previous masks: confidence statistics, mask morphology, temporal consistency, and image-quality measures (brightness, blur, contrast, specular ratio). Features require only the raw frame and segmenter output, and take under 0.5 ms per frame (median 0.48 ms, p_{95} 0.53 ms).

The feature design is intentionally restricted to quantities available outside the segmentation model. Confidence features capture whether the output looks uncertain; shape features capture implausible mask geometry and fragmentation; temporal features capture abrupt changes across adjacent video frames; and image-quality features capture acquisition conditions that can make a confident mask suspect. This makes the monitor compatible with frozen clinical models and avoids relying on architecture-specific internal activations.

The failure-risk model $g_\phi(x_t) \in [0, 1]$ is an XGBoost classifier [30]; learner ablations in the supplement show that the representation is not unique to XGBoost. Training uses clean out-of-fold predictions and corrupted training frames, while threshold calibration is held out from model fitting. At deployment, ground truth is unavailable and an alarm is raised when $g_\phi(x_t) > \lambda$.

The safety objective is miss-rate control: among failed frames, the fraction not alarmed should stay below target α :

$$\text{miss}(\lambda) = \frac{|\{i : g_\phi(x_i) \leq \lambda, \text{fail}(I_i) = 1\}|}{|\{i : \text{fail}(I_i) = 1\}|}. \quad (2)$$

Global split-conformal calibration [22] chooses one threshold λ on a calibration split. Because acquisition severity changes the failure distribution, we also use Mondrian calibration [27]: frames are partitioned into groups g (corruption severity for degraded frames; blur tertile for clean frames), and each group receives

its own threshold

$$\lambda_g = \sup \left\{ \lambda : \frac{|\{i \in \mathcal{G}_g : g_\phi(x_i) \leq \lambda, y_i = 1\}|}{|\{i \in \mathcal{G}_g : y_i = 1\}|} \leq \alpha \right\}. \quad (3)$$

This equalizes empirical conservatism across groups rather than letting easy conditions dominate the average.

The thresholds are calibrated after score learning. Thus discrimination (whether failures rank above correct frames) and safety control (where to set the alarm threshold) are evaluated separately. This separation is important in rare-event settings: a high AUROC monitor can still be operationally burdensome if the threshold required to catch failures creates many frame-level alarms.

4 Experiments

4.1 Setup

We use EndoVis 2017 Instrument Segmentation [31]: ten robotic surgical sequences, with sequences 1–7 for train/calibration and 8–10 for test (1575/450/825 frames). The frozen segmenter is a ResNet34-U-Net fine-tuned on the training sequences. We apply six corruption types at five severity levels (Gaussian blur/noise, motion blur, brightness, contrast, JPEG), yielding 24,750 corrupted test frames. We evaluate $\tau = 0.5$ and $\tau = 0.75$; clean-test failure prevalence is 1.7% and 6.9%, respectively. Baselines are Max-Softmax, Entropy, and a Temporal Heuristic, plus a learned confidence-only monitor in the supplement.

We use four generalization protocols. *Zero-shot* trains the monitor on clean frames only and tests on all corruptions. *LOCO* trains on five corruption types and tests on the held-out sixth type; this is the robustness headline because the test degradation operator is unseen. *Severity extrapolation* trains on severities 1–3 and tests on severities 4–5. *In-distribution* trains and tests across all corruption types and is reported only as an upper bound.

4.2 Clean Test and Generalization

On held-out clean test (Table 1), TCSR-Monitor improves AUROC over all audited baselines at both thresholds. AUPRC is prevalence-sensitive: entropy is stronger at $\tau = 0.5$, whereas TCSR-Monitor gives the best audited AUPRC at $\tau = 0.75$.

LOCO (0.814) is the robustness headline because the test corruption type is absent from training. The small gap to the in-distribution upper bound (0.827) argues against simple corruption fingerprinting, and the monitor beats entropy on all six held-out types.

The clean-test and LOCO results answer different questions. Clean test measures behavior on nominal deployment-like frames, where failures are rare and

Table 1. Held-out clean test results (seed 0; 95% CI from 1000 bootstrap samples where available). AUPRC should be interpreted relative to the rare failure prevalence.

Method	$\tau = 0.5$ AUROC	$\tau = 0.5$ AUPRC	$\tau = 0.75$ AUROC	$\tau = 0.75$ AUPRC
TCSR-Monitor (ours)	0.877	0.468	0.793	0.301
Entropy [5]	0.764	0.635	0.483	0.220
Max-Softmax [3]	0.536	0.087	0.509	0.085
Temporal Heuristic	0.338	0.013	0.460	0.074
Failure prevalence	1.7% (14/825)		6.9% (57/825)	

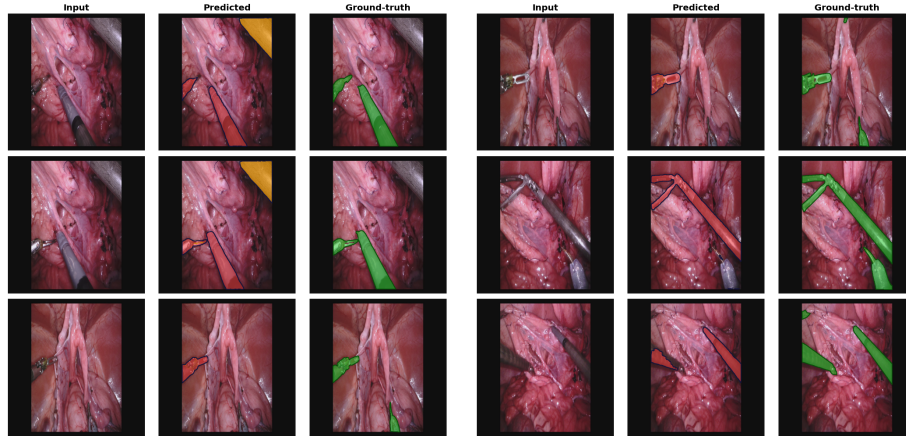


Fig. 2. Qualitative examples of sub-threshold confident failures ($0.5 \leq \text{IoU} < 0.75$). Each triplet shows the input, predicted mask, and ground-truth mask. The segmenter produces plausible-looking but incomplete or over-extended instruments, illustrating why an external failure monitor is needed even when the output appears confident.

AUPRC is prevalence-sensitive. LOCO stresses whether the learned score captures transferable failure patterns under acquisition degradation. The combination is more informative than either setting alone: clean test guards against excessive alarms on normal video, while LOCO guards against memorizing the training corruptions.

4.3 Circularity and Conformal Safety

To test whether TCSR-Monitor predicts *failure* rather than simply *corruption*, we evaluate only corrupted test frames and discriminate correct masks ($\text{IoU} \geq 0.75$) from failed masks ($\text{IoU} < 0.75$). Table 3 shows AUROC 0.946, while entropy is only 0.594. This control is essential because a monitor could otherwise appear successful by detecting low image quality rather than segmentation failure. Restricting the evaluation to corrupted frames removes the easy corrupted-versus-clean shortcut and asks whether the monitor can tell when the segmenter is actually wrong within degraded video.

Table 2. Generalization hierarchy and LOCO breakdown (AUROC). LOCO is the robustness headline; in-distribution is an upper bound.

Protocol / held-out type	Monitor	Entropy	Wins
Zero-shot	0.753	0.465	24/30
LOCO overall	0.814	0.481	26/30
Severity extrapolation	0.831	0.408	8/12
In-distribution upper bound	0.827	0.465	26/30
Brightness	0.801	0.331	4/5
Contrast	0.912	0.839	3/5
Gaussian blur	0.806	0.455	5/5
Gaussian noise	0.811	0.319	4/5
JPEG compression	0.773	0.444	5/5
Motion blur	0.779	0.497	5/5

Table 3. Circularity control and Mondrian conformal calibration. Left: within-corrupted AUROC for failure vs. correct masks. Right: miss-rate by severity at $\alpha = 0.10$.

Severity	Monitor	Entropy	FA-correct	Severity	Global miss	Mondrian miss
1	0.807	0.484	6.5%	1	50.5%	9.3%
2	0.844	0.391	13.3%	2	22.5%	10.1%
3	0.802	0.260	63.7%	3	9.2%	9.9%
4	0.861	0.366	35.8%	4	6.1%	10.0%
5	0.825	0.478	8.9%	5	5.2%	9.9%
Overall	0.946	0.594	12.6%	Overall	10.0%	9.9%

Mondrian calibration fixes the safety imbalance hidden by global calibration: the same overall 10.0% global miss-rate is obtained while missing 50.5% of severity-1 failures. Per-severity thresholds restore near-10% miss-rate for all severities, but at a cost: corrupted false-alarm rate increases from 17.6% to 34.2% overall and from 12.8% to 58.7% at severity 1 (supplement). Thus the claim is safety redistribution, not clinical alarm readiness.

4.4 SAM2 Transfer

We apply the U-Net-trained monitor to SAM2-Tiny [10] run zero-shot with oracle bounding-box prompts. SAM2 fails more often than the fine-tuned U-Net (51.2% vs. 6.9% at $\tau = 0.75$), yet the transferred monitor reaches AUROC 0.930 (Table 4), recovering 93.3% of a SAM2-supervised monitor. Entropy is stronger on SAM2 because these zero-shot failures appear as high uncertainty; the transfer result supports feature portability, not universal superiority over uncertainty baselines.

Table 4. Cross-model transfer to SAM2-Tiny.

Condition	$\tau = 0.5$ AUROC	$\tau = 0.75$ AUROC
U-Net monitor $\rightarrow$ SAM2 (zero-shot transfer)	0.896 [0.873, 0.915]	0.930 [0.912, 0.945]
Entropy $\rightarrow$ SAM2	0.984 [0.975, 0.990]	0.988 [0.979, 0.994]
SAM2-specific monitor (upper bound)	0.997 [0.995, 0.999]	0.997 [0.994, 0.999]
Transfer efficiency (transfer / upper bound)	89.8%	93.3%

5 Discussion

TCSR-Monitor is best understood as a validation-aware monitoring framework, not as a claim that one classifier dominates all baselines. The LOCO and circularity experiments address the two most likely reviewer objections: corruption fingerprinting and corruption detection. The confidence-only, learner, temporal, and sequence controls in the supplement narrow the claim: confidence cues are strong, the full feature set adds modest but positive LOCO gains, and random forest matches or exceeds XGBoost. The novelty is therefore the combination of observable features with protocol-qualified validation and group-conditional safety calibration.

The supplement fills in the main controls behind this interpretation. The learned confidence-only baseline is deliberately stronger than raw entropy or max-softmax because it receives the same learner family as the full monitor. The full feature set improves over it under LOCO, but only modestly, so the claim is not that confidence is irrelevant. Instead, shape, temporal, and image-quality features add incremental generalization signal where confident failures are the target.

The main limitation is scope. All validated discrimination and calibration results use EndoVis 2017 with synthetic corruptions; real multi-site surgical video remains future work. A cached CholecSeg8k audit produced failures for every frame under the current pipeline and therefore could not provide AUROC or AUPRC external validation. SAM2 uses oracle bounding-box prompts and does not replace a second fine-tuned surgical-backbone study. Finally, conformal coverage is empirical rather than certified under temporal dependence, and frame-level alarm rates are too high for direct clinical use without event aggregation or workflow suppression.

These limitations narrow the deployment claim. The present evidence supports TCSR-Monitor as a credible research baseline and validation protocol for monitoring surgical segmentation under acquisition degradation. It does not yet establish a clinically deployable alert stream, nor does it prove universal transfer across hospitals, procedures, or fine-tuned backbone families.

6 Conclusion

We presented TCSR-Monitor, a safety-monitoring framework for surgical segmentation failures under acquisition degradation. By pairing a lightweight feature-

based monitor with LOCO generalization, circularity control, and Mondrian conformal calibration, the paper provides a credible and bounded baseline for detecting confident segmentation failures. The results support feature-based monitoring when fine-tuned segmenters fail silently, while also showing where uncertainty baselines and alarm-burden limitations remain important.

References

1. A. Shvets, A. Rakhlin, A. A. Kalinin, and V. Iglovikov, “Automatic instrument segmentation in robot-assisted surgery using deep learning,” in *Proc. IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, 2018.
2. M. Allan *et al.*, “2017 robotic instrument segmentation challenge,” *arXiv:1902.06426*, 2019.
3. D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017.
4. D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
5. A. Malinin and M. Gales, “Predictive uncertainty estimation via prior networks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2018.
6. C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017.
7. O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, 2015.
8. B. Dong, W. Wang, D.-P. Fan, J. Li, H. Fu, and L. Shao, “Polyp-PVT: Polyp segmentation with pyramid vision transformers,” *arXiv:2108.06932*, 2021.
9. A. Kirillov *et al.*, “Segment anything,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023.
10. N. Ravi *et al.*, “SAM 2: Segment anything in images and videos,” *arXiv:2408.00714*, 2024.
11. Y. Ding *et al.*, “SegSTRONG-C: Segmenting surgical tools robustly on non-adversarial generated corruptions,” *arXiv:2407.11906*, 2024.
12. C. Shen *et al.*, “Surgical SAM 2: Real-time segment anything in surgical video by efficient frame pruning,” *arXiv:2408.07931*, 2024.
13. U. Michieli and P. Zanuttigh, “Continual semantic segmentation via repulsion-attraction of sparse and disentangled latent representations,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2021.
14. B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.
15. K. Lee, K. Lee, H. Lee, and J. Shin, “A simple unified framework for detecting out-of-distribution samples and adversarial attacks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2018.
16. F. Granese *et al.*, “DOCTOR: A simple method for detecting misclassification errors,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.
17. C. Corbière *et al.*, “Addressing failure prediction by learning model confidence,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.

18. A. Jungo and M. Reyes, “Assessing reliability and challenges of uncertainty estimations for medical image segmentation,” *arXiv:1907.03338*, 2019.
19. A. Mehrtaash *et al.*, “Confidence calibration and predictive uncertainty estimation for deep medical image segmentation,” *IEEE Trans. Med. Imaging*, 2020.
20. M. Rottmann, P. Colling, T. P. Hack, R. Chan, F. Hüger, P. Schlicht, and H. Gottschalk, “Prediction error meta-classification in semantic segmentation: Detection via aggregated dispersion measures of softmax probabilities,” *arXiv:1811.00648*, 2018.
21. S. K. Jung *et al.*, “FSNet: A failure detection framework for semantic segmentation,” *arXiv:2108.08748*, 2021.
22. V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.
23. A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” *arXiv:2107.07511*, 2021.
24. Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.
25. Y. Geifman and R. El-Yaniv, “SelectiveNet: A deep neural network with an integrated reject option,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019.
26. A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, “Conformal risk control,” *arXiv:2208.02814*, 2022.
27. V. Vovk, D. Lindsay, I. Nouretdinov, and A. Gammerman, “Mondrian confidence machines / conformal predictors,” *Symp. Learn. Data Sci. (SLDS)*, 2003.
28. C. Lu *et al.*, “Improving trustworthiness of AI disease severity rating in medical imaging with ordinal conformal prediction sets,” in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, 2022.
29. A. N. Angelopoulos *et al.*, “Image-to-image regression with distribution-free uncertainty quantification and applications in imaging,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022.
30. T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, 2016.
31. S. Bodenstedt *et al.*, “Comparative evaluation of instrument segmentation and tracking methods in minimally invasive surgery,” *arXiv:1805.02475*, 2018.
32. W.-Y. Hong *et al.*, “CholecSeg8k: A semantic segmentation dataset for laparoscopic cholecystectomy based on Cholec80,” *arXiv:2012.12453*, 2020.